The Hungry Bird Robot

Time: 35–40 min · **After:** Module 1 beak races (spoon, tweezers, fork)

> **Learning goal (students):** A robot only knows the examples we give it. If the examples are incomplete, its rule can be wrong.

LEARNING POINTS (TEACHERS)

1. **AI = a rule from examples** — not magic.
2. **Test on something new** — a mystery bird the robot hasn't "seen."
3. **Bias can mean incomplete data** — spoon-only sample skews the lesson.
4. **Humans are responsible** for the lesson the AI got.

Do not invent a silly rule just to tear it down. If a student offers a looks/color shortcut, you may test it as an optional add-on only.

LESSON ARC

1. Hungry-bird question + fair rule from scores
2. Test on a mystery bird (Robot follows the card)
3. Unfair sample (only spoon data) → rule breaks
4. Ask: "Did the robot fail, or did we give it a bad / incomplete lesson?"

WHAT TO SAY — STEP BY STEP

Hook (2 min)

> *"Yesterday we were birds hunting for food. Today you'll build a tiny AI — a robot that decides where a hungry bird should go for the most food: a wet place or a forest. The robot is not smart by itself. It only follows a rule we write from numbers."*

Optional: choose one volunteer as the Robot. They may only read their index card later.

1) Fair rule (8–10 min)

> *"If you were a robot, and you could only look at these numbers, what rule would you use to decide where the hungry bird should go for the most food — a wet place or a forest?"*

Beak	Seeds	Worms	Water food
Spoon	6	2	14
Tweezers	3	15	1
Fork	11	8	2

Students may say: "High water → wet," "High worms → forest," "Spoons go wet."

Guide toward one short rule, e.g. *If Water food ≥ 10 → Wet place. Otherwise → Forest.* Write it on the Robot card.

> *"This rule uses the race evidence. That's a fair lesson for a robot."*

2) Test mystery bird (8 min)

> *"Here's a new bird. No vibes. Robot: only use the card."*

Example A: Water food 16, Worms 3 → Wet. Example B: Water food 2, Worms 18 → Forest.

> *"Did the robot know birds? No. It matched numbers to a rule. That's all AI is doing here."*

3) Unfair sample (10–12 min)

> *"Plot twist. Pretend scientists only studied spoons. Cover tweezers and fork. Using ONLY this, what rule would the robot learn?"*

Students may say: "Always wet," "Spoons are best," "Water food is all that matters."

Uncover other beaks; run spoon-only rule on tweezers — it fails.

> *"The robot isn't broken. We gave it an incomplete lesson. That's biased data — an unfair sample."*

> *"Same idea if an AI only sees one neighborhood or one group of people — then its rules fail for everyone else."*

4) Close (3–5 min)

> *"Did the robot fail, or did we give it a bad / incomplete lesson?"*

Exit ticket: "Today the AI messed up because ____."

TRANSITIONS (KEEP IT NATURAL)

From → To	Say
Hook → Fair rule	“Robots need a rule made from evidence. Let’s build one.”
Fair rule → Test	“A rule isn’t real until we try it on a new bird.”
Test → Unfair sample	“It worked with everyone’s data. What if we hid most of the data?”
Unfair sample → Close	“Who made the mistake — the robot, or the teachers of the robot?”

IF STUDENTS SAY…

Student	You
“AI is magic / smarter than us”	“Point to the card. That’s the whole brain today.”
Long complicated rule	“Pick one number the robot can check in 2 seconds.”
Blames the Robot student	“The person isn’t wrong — the lesson we wrote is.”
“Just use common sense”	“Robots don’t get common sense unless we put it in the data/rule.”
Offers “yellow / pretty / metal” rule	Park it; optionally test later — don’t force a silly-rule skit.

SUCCESS LOOKS LIKE

A sixth grader can say: **“We can trick an AI by only showing it some results — the robot failed because of our incomplete lesson.”**

MATERIALS CHECKLIST

- Module 1 scores on board (or sample table on lesson page)
- Student rule cards
- Robot index card
- Something to cover tweezers/fork rows